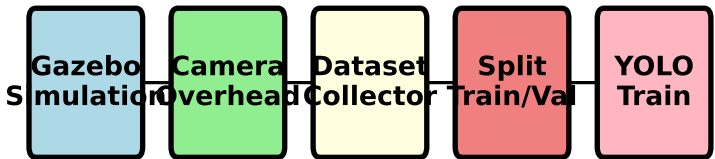
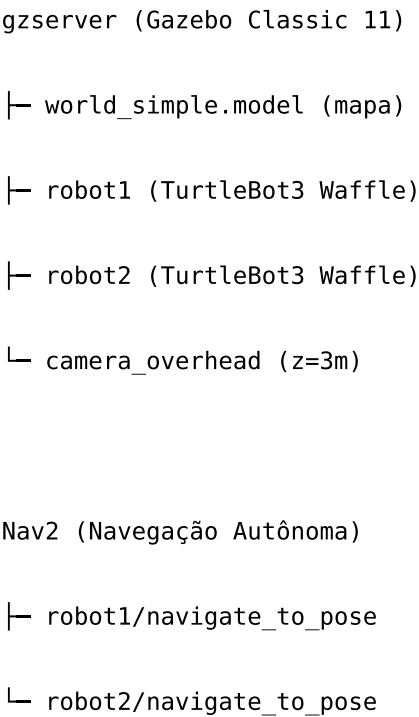


CERISE Pipeline

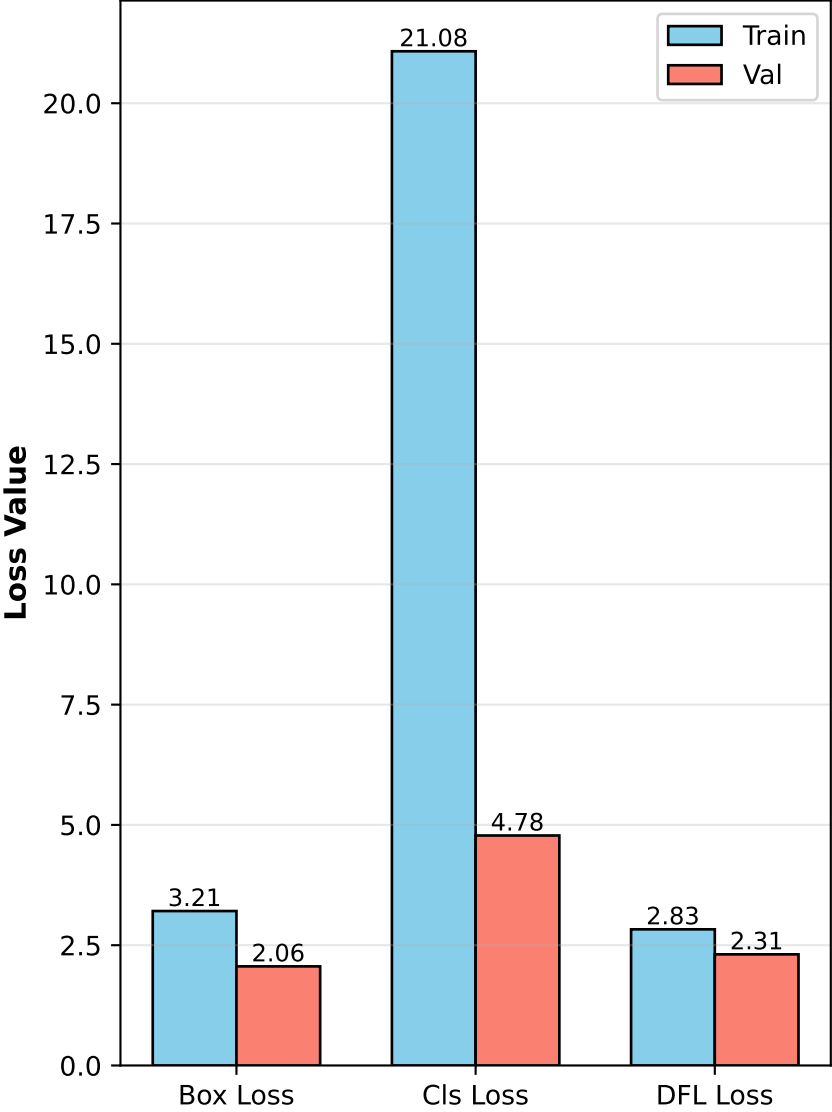


- Gazebo com mapa
- 2 robôs TurtleBot3
- World_simple.model
- ⚠ **Câmera não publicou**
- Dataset dummy criado
- YOLO treinado (1 época)

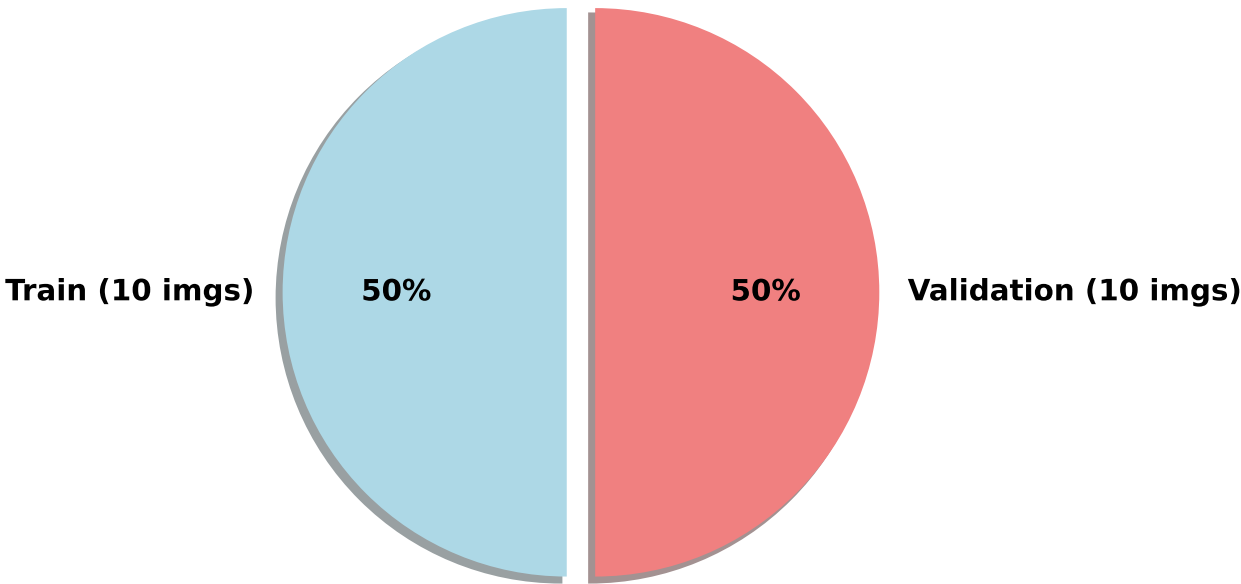
System Architecture



YOLO Training Metrics (Epoch 1)



Dataset Split (80/20)



YOLOv8n Model

Parameters: 3,011,043
Layers: 130
GFLOPs: 8.2

Training Config:

- Epochs: 1
- Image Size: 640x480
- Batch Size: 16
- Optimizer: AdamW

Results:

- mAP50: 0%
- mAP50-95: 0%
- (Expected com dataset real)

Próximas Etapas

- Coletar dataset real** *60+ frames*
- Corrigir plugin câmera** *Arquivo ros_camera*
- Treinar com dados reais** *50 epochs*
- Inferência tempo real** *Execução online*
- Validar acurácia** *YOLO vs AMCL*
- Publicar resultados** *ICRS 2025*